Tactix no-MAB stabilization milestone and remaining correctness work

Report to Andrew

28 September 2026

Decision. The user accepted the no-MAB Tactix stabilization milestone on 28 September 2026, explicitly satisfied with solver performance whether or not the last WV100 proof check succeeds. The accepted scope is the four fixed regression sets and the disclosed correctness exceptions below. This is not a claim of universal N1–N4, full WV2 coverage, or Tier D competitiveness. The solver remains the immutable no-MAB source `d170b4a5`, Release SHA-256 beginning `2fc7421f`; no public X7 action was narrowed. Its native DSR proof route now uses validated DSR-to-DRAT, drat-trim and CakeLPR for witness-free streams; witnessed streams use no-option dsr-trim and Trestle. Startup PBP uses VeriPB and CakePB. Every accepted SAT model and UNSAT proof is checked against the original CNF.

Measured four-set evidence. The new hybrid-checker reruns use exactly the fixed AM519 instance lists and unchanged solver binary. Independent scoring of the three terminal sets gives:

Set	Checked	Models	Proofs	Solver seconds	Remaining coverage
Rank 1–50	50/50	25	25	7,658.828	none
B28+C5	33/33	17	16	960.046	none
WV2–89	79/89	48	31	4,047.450	10 solver timeouts
WV100	99/100	33	66	2,747.495	1 proof still checking

Rank and B28+C5 are below the user's five-percent aggregate ceilings of 8,521.522 and 999.876 seconds, respectively. Every Rank/B28+C5 case finished within its 1,200-second solver limit; the 79 decided WV2 cases finished within 600 seconds. Rank 22, the prior checker bottleneck, now has an independently verified drat-trim/CakeLPR UNSAT certificate. WV100's last `ncc` checker was running at this report snapshot. Its 99 finished records are independently verified; no checker rejection or timeout had occurred. The pinned four-set scorer `1739144` waits for the final WV100 record and will record its terminal disposition separately.

Correctness status still open. There is no pending fuzz job. The terminal Claude Code Opus AM531 no-MAB campaign ran 23,300 fresh cases and found zero solver crashes, invalid models, rejected produced proofs, checker-route mismatches, or parity failures. Its *frozen acceptance label is still NOT PASSED*: nine truncation negative controls were inapplicable because their retained prefixes already refuted the CNF; corrected uncounted controls rejected 9/9. The targeted legal-public-action D2F analogue was not reached. A historical direct-Proof-API D2F counterexample does not establish a reachable public X7 failure, but public reach has not been disproved. The user accepted these disclosed gaps for this milestone. Independent hybrid-route positive and negative tests, compute-node validation, and the benchmark certificates cover the checker change; no new solver fuzz was needed for unchanged solver bytes. If the remaining WV100 proof fails, that is a separate N2 investigation, not a change to the already accepted performance measurement.

Full Tier D and learning readiness. The four-winner comparison's retained Kissat-sc2024, LymphoSAT, and AE-Kissat-MAB arms continue in array `1733043`. Corrected Satsuma+Kissat hybrid array `1737781` starts after that winner array, followed by no-MAB Tactix hybrid array `1737785`; both contain all 1,166 Tier D instances. These `cpu-q` campaigns use four CPUs, 32 GiB, 5,000 seconds for solving and four hours for independent proof checking per task, with at most 120 benchmark tasks running, anonymous scratch cleanup, and ten-minute 700 GB pending-only storage guards. Dependent scorer `1739244` will compare the five arms after the last array. No Tier D performance claim is available yet.

The code supports typed, ordered and dosed explicit inprocessing actions and a configurable default schedule. It is *not ready* for a learner that switches inprocessing, branching and deletion strategies during a solve. The mature default inprocessing scheduler owns separate slot state; branch and reduction subpolicies are borrowed pre-solve pointers without safe live-switch boundaries or state migration; reduction controls a prefix of an already-ranked candidate list. The next session should consume WV100 and Tier D outcomes, preserve the no-MAB default, then design owned switchable policy state, deterministic observations and reward attribution before learning experiments.

Evidence and next action. See `benchmarks/reports/2026-09-28-am551-no-mab-hybrid-milestone-accepted.md`, `benchmarks/runs/20260928-am531-no-mab-opus/REPORT.md`, `benchmarks/reports/2026-09-28-am533-learning-readiness-code-audit.md`, and `PROGRESS.md`. Consume WV100's final proof and scorer without changing checker contract; investigate any failed certificate. Complete and score the queued Tier D campaign, then assess N3 and the broader N1–N4 gaps separately.